

Literature Review Report

Summary

Recent advances in computer vision have significantly improved robotic perception through deep neural networks. However, these models require substantial computational resources, limiting deployment on embedded robotic platforms. Knowledge Distillation (KD) has emerged as an effective model compression technique, while Quantum Machine Learning (QML) offers a new paradigm for improving learning using variational quantum circuits. Current research indicates promising results for hybrid quantum-classical models, although practical deployment on edge devices remains challenging. The literature also reveals a significant research gap: very few studies combine Knowledge Distillation with Quantum Machine Learning for lightweight robotic vision, making this project both novel and timely

No.	Reference	Key Findings	Relevance to Proposed Work
1	Habib et al. (2024), <i>A Comprehensive Review of Knowledge Distillation in Computer Vision</i>	Reviews response-based, feature-based and relation-based KD techniques and highlights KD as the leading model compression method for resource-constrained vision systems.	Establishes KD as the foundation of the proposed framework. (arXiv)
2	Peral-García et al. (2024), <i>Systematic Literature Review: Quantum Machine Learning and its Applications</i> , Computer Science Review	Reviews 94 QML studies, showing image classification as one of the most active application areas while identifying hardware limitations of current quantum systems.	Provides the QML background and identifies research gaps. (ScienceDirect)
3	Liang et al. (2024), <i>Design and Analysis of Quantum</i>	Surveys quantum neural networks, QSVMs,	Supports the choice of variational quantum

	<i>Machine Learning: A Survey</i> , Connection Science	variational circuits and discusses practical implementation challenges.	circuits. (Taylor & Francis Online)
4	Lu et al. (2024), <i>Quantum Machine Learning: Classifications, Challenges, and Solutions</i> , Journal of Industrial Information Integration	Discusses algorithm categories, noise mitigation, scalability and future directions for practical QML.	Highlights current limitations addressed by hybrid models. (ScienceDirect)
5	Chen (2024), <i>A Novel Image Classification Framework Based on Variational Quantum Algorithms</i> , Quantum Information Processing	Demonstrates improved image classification using variational quantum circuits integrated with classical networks.	Directly related to the proposed hybrid quantum student model. (Springer)
6	Zeguendry et al. (2023), <i>Quantum Machine Learning: A Review and Case Studies</i> , Entropy	Compares QSVM, Quantvolutional CNNs and Variational Quantum Classifiers using benchmark datasets.	Provides implementation guidance for hybrid quantum classifiers. (MDPI)
7	Kharsa et al. (2023), <i>Advances in Quantum Machine Learning and Deep Learning for Image Classification: A Survey</i> , Neurocomputing	Reviews quantum image classification methods and identifies challenges in encoding RGB images and scaling QML.	Supports the computer vision component of the project. (ScienceDirect)
8	Cerezo et al. (2022), <i>Challenges and Opportunities in Quantum Machine Learning</i> , Nature Computational Science	Discusses trainability, barren plateaus, quantum neural networks and prospects for quantum advantage.	Explains limitations considered in the proposed framework. (Nature)

9	Li & Deng (2021), <i>Recent Advances for Quantum Classifiers</i> , Science China Physics, Mechanics & Astronomy	Reviews variational quantum classifiers, quantum kernels and adversarial robustness.	Supports selection of VQC architecture. (arXiv)
10	Mangini (2023), <i>Variational Quantum Algorithms for Machine Learning: Theory and Applications</i>	Presents theoretical foundations and applications of variational quantum circuits in classification.	Provides mathematical basis for the quantum component. (arXiv)
11	Blance & Spannowsky (2021), <i>Quantum Machine Learning for Particle Physics Using a Variational Quantum Classifier</i> , JHEP	Demonstrates practical hybrid optimization using VQCs on NISQ devices.	Validates feasibility of hybrid quantum-classical classifiers. (Springer)
12	IBM Patent US20240070508A1 (2024), <i>Quantum Variational Network Classifier</i>	Introduces a variational quantum classifier capable of learning nonlinear decision boundaries for classification.	Shows industrial interest in quantum classifiers similar to the proposed work. (Google Patents)

Research Gap

The literature reveals several important gaps:

- Knowledge Distillation research is mature for computer vision but remains almost entirely classical.
- Quantum Machine Learning research primarily investigates standalone quantum classifiers rather than model compression.
- Existing hybrid quantum-classical image classification studies focus on improving classification accuracy rather than reducing computational complexity for edge deployment.

- Very limited work integrates Knowledge Distillation, Variational Quantum Circuits, and edge-based robotic vision into a unified framework, representing a clear opportunity for further research.

Conclusion

The reviewed literature demonstrates that Knowledge Distillation is an effective approach for model compression, while Quantum Machine Learning provides promising methods for enhancing feature representation and classification. However, the integration of quantum-assisted knowledge distillation for lightweight robotic vision remains largely unexplored. The proposed project addresses this gap by combining a high-capacity teacher network, a compact student CNN, and a variational quantum circuit to produce an efficient hybrid model suitable for deployment on resource-constrained edge devices such as Raspberry Pi.